

Ghanshyam Gupta

Delhi, Delhi

ghanshyamg1402@gmail.com

+918299268243

Professional Summary results-driven electronics engineering leader with 10+ years of hands-on experience spanning medical-robotics, automotive forklift controls, railway signalling, and industrial electronics. Expert in PCB , firmware, system integration, and rigorous quality assurance (IQC/OQC/PQC, CAPA, PFMEA). Proven track record of delivering complex projects on time, reducing defect rates, and improving production efficiency. Adept at cross-functional collaboration, root-cause analysis, and mentoring teams to achieve high-quality outcomes. Seeking a engineering managerial or technical-lead role where I can drive innovation, ensure compliance, and elevate product reliability.

Willing to relocate: Anywhere

Work Experience

Technical Expert/Technical Advisor/Technical Consultant

IFCI Limited-Delhi

Full-time

June 2024 to Present • Full-time

- IFCI (Industrial Finance Corporation of India) is under the ownership and jurisdiction of the Ministry of Finance, Government of India, specifically falling under the Department of Financial Services (DFS), which handles its oversight and administration as a key development finance institution.
- IFCI Ltd. acts as the Project Management Agency (PMA) for the Ministry of Electronics and Information Technology (MeitY) for schemes promoting the manufacturing of electronics components and semiconductors in India.
- I'm working with IFCI as a Domain Expert for the MeitY PLI Electronics Scheme.
- Provided technical expertise across electronics components, harness, PCB fabrication, and PCB assembly industries, ensuring optimal manufacturing processes.
- Delivered technical guidance for IT-hardware manufacturing (laptops, AIOs, USSF, servers), aligning production with industry standards.
- Offered consultancy services to electronic-components firms, focusing on process optimization, technology selection, and design verification of machinery.
- Conducted detailed analysis and review of capital-expenditure (Capex) items, including machinery, utilities, and technology required for electronics manufacturing.
- Classified Capex items under appropriate asset heads—Plant & Machinery, Utilities, R&D, and Technology Deployed—ensuring accurate accounting and reporting.
- Developed a comprehensive understanding of electronic items and components, categorizing them into broad classifications for streamlined procurement.
- Provided technical know-how for planning, designing, and verifying production machinery and ancillary equipment in electronics plants.
- Identified and verified the critical equipment and plant-machinery requirements essential to the electronics manufacturing workflow.
- Reviewed Capex proposals, confirming alignment with production needs and recommending appropriate classifications.
- Verified installation of Capex assets, assessed commercial production readiness, and monitored related operational activities.
- Executed project inspections, investment verifications, and related audits to ensure compliance and performance.
- Contributed technical expertise to the implementation of the Production-Linked Incentive (PLI) scheme for electronics manufacturing, guiding policy design and execution.

- Analyzed industry trends, market research, and policy frameworks to inform PLI scheme structures and eligibility criteria.
- Collaborated with industry leaders, research institutions, and government agencies to refine scheme guidelines and standards.
- Developed technical guidelines, evaluation criteria, and standards for PLI scheme participants, ensuring transparent selection processes.
- Conducted compliance audits for electronics and IT-hardware firms under the PLI scheme, assessing production, investment, and employment metrics.
- Identified improvement opportunities and provided actionable recommendations to enhance scheme effectiveness and participant performance.
- Managed government-compliance projects, ensuring electronics manufacturers meet all statutory and regulatory requirements.
- Served as technical consultant to the Ministry of Electronics & Information Technology, providing expert advice on policy implementation and scheme eligibility.
- Delivered technical advisory services to a nationwide portfolio of electronics manufacturing firms, covering PCB, EMS, and IT-hardware production.
- Conducted compliance audits across India-based electronics companies, verifying capital investment, sales performance, and employment data.
- Met with Managing Directors and Directors of audited firms to review progress, discuss findings, and recommend corrective actions.
- Verified Capex investments, sales figures, and headcount during audits to confirm adherence to incentive scheme criteria.
- Monitored production lines (SMT, FATP, etc.) to ensure operational compliance and identify any gaps in manufacturing processes.
- Reviewed critical documents such as lease agreements, vendor contracts, customer agreements, and environmental/pollution permits.
- Collected and analyzed production capacity, product specifications, raw-material usage, and contract-manufacturing data for EMS clients.
- Provided the IT team with technical inputs on portal-related issues and required system changes, enhancing data accuracy and reporting.
- Offered technical consultation to Ministry of Electronics and PLI-eligible companies on LSEM and PLI 2.0 schemes for IT hardware.
- Prepared technical recommendations for MeitY based on queries received under PLI schemes, supporting policy decision-making.
- Performed sales and Capex register verification, cross-checking entries with supporting invoices and financial documents.
- Conducted physical inspections and on-site verifications to confirm asset existence and operational readiness.
- Handled miscellaneous MIS tasks, invoice appropriation, payment requests for concurrent auditor fees, and verification fee invoicing for MeitY.
- Ensured adherence to digital-system directives, contributed to process improvements, and mentored junior staff.
- Provided feedback on key organizational matters and actively participated in internal committees, company events, national festivals, and other government initiatives.

Manager

SS Innovations Pvt Ltd-Gurgaon, Haryana

Full-time

September 2022 to June 2024 • Full-time

- Lead the design, development, verification and validation of next-generation minimally invasive surgical robots (Mantra) for a robotics-focused medical equipment manufacturer.
- Apply SolidWorks for 3-D modeling, mechanical layout and assembly of robotic subsystems.
- Manage all wire and cable assemblies, ensuring proper routing, termination and compliance with EMC requirements.
- Supervise PCB testing and troubleshooting, including functional verification and fault isolation.

- Specify and qualify EtherCAT cables and shielding to support high-speed motion control networks.
- Design and integrate BLDC and brushed motor solutions, performing motor selection, sizing and tuning.
- Program and configure microcontrollers (STM32, Arduino) and serial communication interfaces (RS-422, RS-232).
- Conduct root-cause analysis on electrical/electronic failures and implement corrective actions.
- Design, build and commission custom test rigs and validation equipment for robotic components.
- Install, commission and service robotic systems on-site, ensuring safe and reliable operation.
- Configure encoders, motion controllers and actuators; tune drives (SSI, ELMO, Technosoft, Ingenia Denali, etc.) and update firmware as needed.
- Develop harness designs and wiring diagrams aligned with system functional and performance specifications.
- Create accurate Bills of Material (BOM) and maintain component databases in ERP/SAP.
- Lead procurement of electronic parts, components and accessories, negotiating with suppliers for cost and lead-time optimization.
- Generate PCB checklists, quality control documents and traceability records to support production.
- Translate user requirements into detailed hardware specifications and design constraints.
- Implement data-acquisition and analysis methods to evaluate design performance and validate safety protocols.
- Execute test protocols and verification steps to demonstrate system safety and regulatory compliance.
- Troubleshoot complex electrical and electronic issues across the product lifecycle, reducing downtime and improving yield.
- Collaborate with cross-functional teams (R&D, manufacturing, QA) to drive continuous improvement and meet project milestones.

Senior Engineer

Watrana Traction Pvt Ltd-Greater Noida, Uttar Pradesh

Full-time

November 2021 to September 2022 • Full-time

- Provide end-to-end electronic circuit debugging and troubleshooting for forklift control boards and vehicle-level systems.
- Test and certify battery packs and chargers, ensuring compliance with safety and performance standards.
- Apply knowledge of universal, induction and motor-encoder systems to diagnose and repair drive-train electronics.
- Manage battery-bank configurations, monitoring voltage, current and thermal profiles for optimal operation.
- Service SMPS and UPS units, performing component-level repair and preventive maintenance.
- Verify PCB signal integrity and transmission-line circuitry using oscilloscopes and impedance testing.
- Select appropriate analog and digital components for PCB refurbishment, considering cost, availability and performance.
- Perform reverse-engineering of legacy forklift PCBs to recreate schematics and BOMs for obsolescence management.
- Conduct circuit simulation (DC/AC analysis) to validate design changes before implementation.
- Coordinate with PCB fabricators, assembly houses and component distributors to ensure timely delivery and quality.
- Manage component selection, including semiconductors, resistors and capacitors, while tracking obsolescence and maintaining technical documentation.
- Maintain a centralized online database of repaired PCBA cards, tracking test results, repair history and failure modes.
- Repair a wide range of PCBA cards and controllers used in forklift brands such as Nichiyu, Ace, Bendi, GE, Mitsubishi, Toyota, Danaher, Curtis, BT and Zapi.

- Serve as the second-level technical lead in the R&D department, overseeing all electronic and electrical projects.
- Generate and update technical records, test reports and data sheets for analysis and continuous improvement.
- Ensure adherence to manufacturing processes for bare PCB and PCBA modules, optimizing yield and reliability.
- Collaborate with cross-functional teams to troubleshoot complex issues, reduce downtime and improve product reliability.
- Contribute to process improvements, cost-reduction initiatives and best-practice sharing across the supply-chain network.

Deputy Engineer

Central Electronics Limited-Ghaziabad, Uttar Pradesh

Full-time

July 2014 to November 2021 • Full-time

- Developed and implemented detailed IQC, OQC, and PQC guidelines and inspection criteria for all products.
- Conducted incoming, in-process, and outgoing quality inspections to identify defects and ensure compliance with customer specifications and sampling procedures.
- Tested and repaired SSDAC, HASSDAC, and MSDAC cards/units in the Railway Testing Division, including fault finding, installation, testing, and commissioning of signaling installations.
- Prepared and presented defect analysis reports using CAPA, PFMEA, 5-Why, and Fishbone diagrams; tracked corrective actions and preventive measures.
- Set, monitored, and reported quality assurance goals and KPIs; collaborated with internal teams to improve product and service quality, enhancing customer satisfaction.
- Recorded, maintained, and executed calibration of various instruments per schedule and due dates; ensured traceability and minimized downtime.
- Conducted scheduled internal audits of functions and processes; ensured adherence to 5S guidelines and ISO-9001 standards.
- Organized daily, weekly, and monthly quality meetings; documented incidents, fixes, and improvements in the Quality Management System (QMS).
- Maintained documentation for RDSO, RITES, BIS, and other regulatory inspections.
- Resolved customer feedback and reviews; escalated issues to the improvement team and tracked closure.
- Implemented Value Stream Mapping elements to streamline customer experience and align with business processes.
- Handled customer relationships, providing technical support and after-sales service.
- Demonstrated hard work, punctuality, realism, patience, creativity, positive attitude, and strong teamwork.
- Quick learner with complex problem-solving skills, methodical mindset, critical thinking, and strong interest in technology and mathematics.

Education

Electrical & Electronics Engineering (B.Tech/B.E.)

Dr. A. P. J. Abdul Kalam Technical University-Lucknow, Uttar Pradesh

Full-time | July 2010 to June 2014